

Lab10

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PDB STATISTICS

Q1: What percentage of structures in the PDB are solved by X-Ray and Electron Microscopy.

A1. X-Ray: about 80.95077 and EM: about 12.83843.

```
pdb <- read.csv("pdb_stats.csv")
```

```
#Column names  
str(pdb)
```

```
'data.frame':  6 obs. of  9 variables:  
 $ Molecular.Type   : chr  "Protein (only)" "Protein/Oligosaccharide" "Protein/NA" "Nucleic a  
 $ X.ray            : int  178795 10363 9106 3132 175 11  
 $ EM               : int  21825 3564 6335 221 25 0  
 $ NMR              : int  12773 34 287 1566 33 6  
 $ Integrative      : int  343 8 24 3 4 0  
 $ Multiple.methods : int  226 11 7 15 0 1  
 $ Neutron          : int  84 1 0 3 0 0  
 $ Other            : int  32 0 0 1 0 4  
 $ Total            : int  214078 13981 15759 4941 237 22
```

```
total <- sum(pdb$Total)  
xray <- sum(pdb$X.ray)  
em <- sum(pdb$EM)  
  
xray1 <- xray / total * 100  
em1 <- em / total * 100  
  
xray1
```

[1] 80.95077

em1

[1] 12.83843

Q2. What proportion of structures in the PDB are protein?

A2. About 97.91%

```
protein <- sum(pdb$Total[1:3])
total <- sum(pdb$Total)

protein_prop <- protein / total
protein_prop
```

[1] 0.979118

Q3: Type HIV in the PDB website search box on the home page and determine how many HIV-1 protease structures are in the current PDB?

A3. About 1,147.

Q4. Water molecules normally have 3 atoms. Why do we see just one atom per water molecule in this structure?

A4. It could be due to a x-ray diffraction limitation. Or due to size of the atoms: hydrogens are small and thus not as visible, but the oxygen atom is larger and thus easier to detect.

Q5. There is a critical “conserved water molecule in the binding site. Can you identify this water molecule? What residue number does this water molecule have?

A5. Yes. It is between the two ASP 25 residues, where it helps stabilize ligand binding. Residue number is HOH 308.

Q6. Generate and save a figure clearly showing the two distinct chains of HIV-protease along with the ligand. You might also consider showing the catalytic residues ASP 25 in each chain and the critical water (we recommend “Ball & Stick” for these side-chains). Add this figure to your Quarto document.

A6. *Done below*

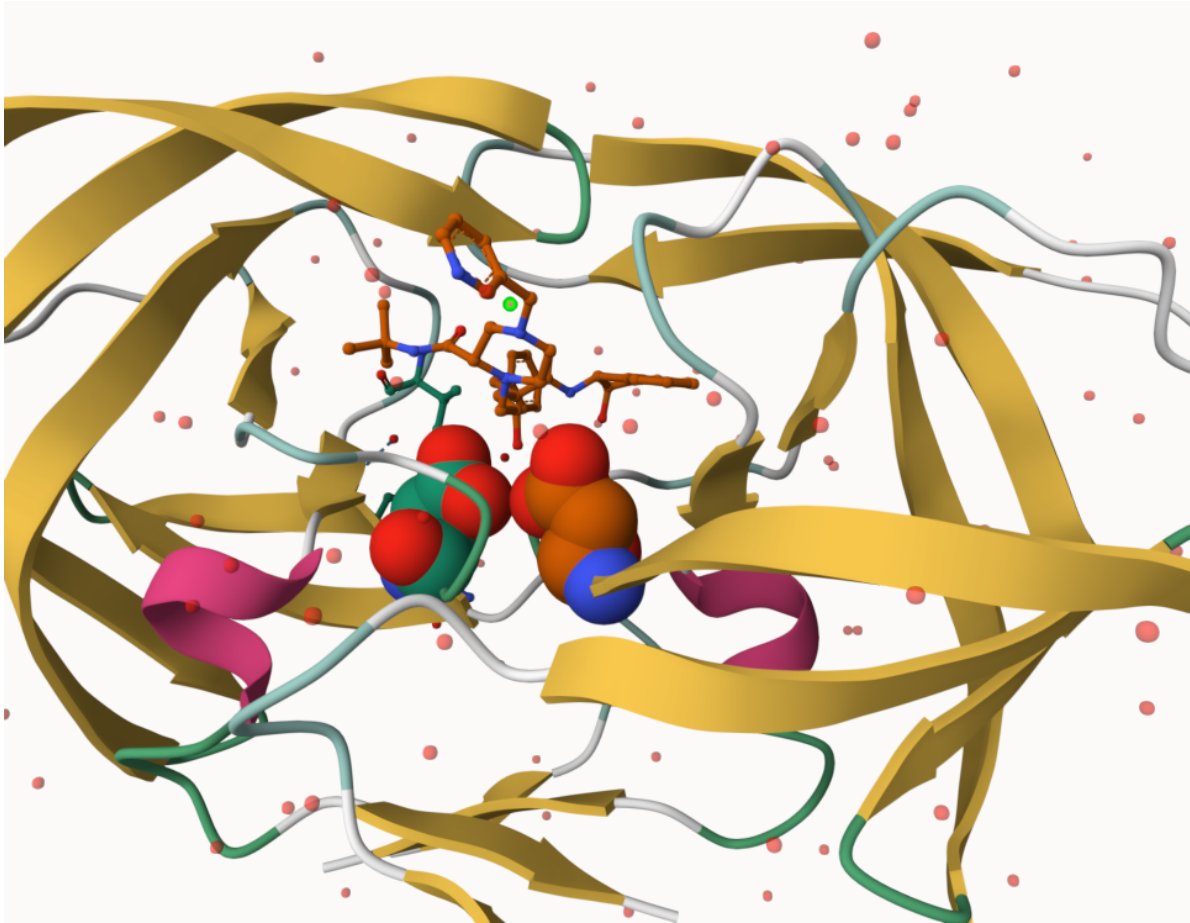


Figure 1: 1HSG ASP 25

INTRODUCTION TO BIO3D IN R

```
library(bio3d)  
pdb <- read.pdb("1hsg")
```

Note: Accessing on-line PDB file

```
pdb
```

Call: `read.pdb(file = "1hsg")`

```

Total Models#: 1
Total Atoms#: 1686, XYZs#: 5058 Chains#: 2 (values: A B)

Protein Atoms#: 1514 (residues/Calpha atoms#: 198)
Nucleic acid Atoms#: 0 (residues/phosphate atoms#: 0)

Non-protein/nucleic Atoms#: 172 (residues: 128)
Non-protein/nucleic resid values: [ HOH (127), MK1 (1) ]

```

```

Protein sequence:
PQITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGIGGFIKVRQYD
QILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNFPQITLWQRPLVTIKIGGQLKE
ALLDTGADDTVLEEMSLPGRWKPKMIGGIGGFIKVRQYDQILIEICGHKAIGTVLVGPTP
VNIIGRNLLTQIGCTLNF

```

```

+ attr: atom, xyz, seqres, helix, sheet,
      calpha, remark, call

```

Q7. How many amino acid residues are there in this pdb object?

198

Q8. Name one of the two non-protein residues.

MK1

Q9. How many protein chains are in this structure?

2

```
attributes(pdb)
```

```

$names
[1] "atom"  "xyz"   "seqres" "helix" "sheet" "calpha" "remark" "call"

```

```

$class
[1] "pdb" "sse"

```

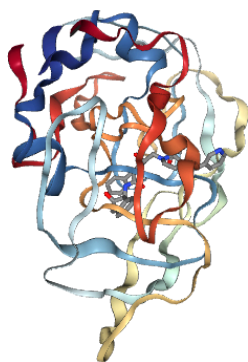
```
head(pdb$atom)
```

	type	eleno	elety	alt	resid	chain	resno	insert	x	y	z	o	b
1	ATOM	1	N	<NA>	PRO	A	1	<NA>	29.361	39.686	5.862	1	38.10
2	ATOM	2	CA	<NA>	PRO	A	1	<NA>	30.307	38.663	5.319	1	40.62
3	ATOM	3	C	<NA>	PRO	A	1	<NA>	29.760	38.071	4.022	1	42.64
4	ATOM	4	O	<NA>	PRO	A	1	<NA>	28.600	38.302	3.676	1	43.40
5	ATOM	5	CB	<NA>	PRO	A	1	<NA>	30.508	37.541	6.342	1	37.87
6	ATOM	6	CG	<NA>	PRO	A	1	<NA>	29.296	37.591	7.162	1	38.40

	segid	elesy	charge
1	<NA>	N	<NA>
2	<NA>	C	<NA>
3	<NA>	C	<NA>
4	<NA>	O	<NA>
5	<NA>	C	<NA>
6	<NA>	C	<NA>

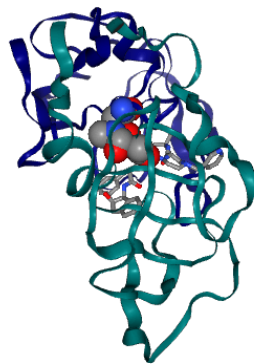
```
library(bio3dview)
library(NGLViewerR)

view.pdb(pdb) |>
  setSpin()
```



```
#Select the important ASP 25 reside
sele <- atom.select(pdb, resno=25)

#and highlight them in spacefill representation
view.pdb(pdb, cols=c("navy", "teal"),
          highlight = sele,
          highlight.style = "spacefill") |>
setRock()
```



PREDICTING FUNCTIONAL MOTIONS OF A SINGLE STRUCTURE

```
adk <- read.pdb("6s36")
```

Note: Accessing on-line PDB file
PDB has ALT records, taking A only, rm.alt=TRUE

```
adk
```

```
Call: read.pdb(file = "6s36")
```

```
Total Models#: 1
```

```
Total Atoms#: 1898, XYZs#: 5694 Chains#: 1 (values: A)
```

```
Protein Atoms#: 1654 (residues/Calpha atoms#: 214)
```

```
Nucleic acid Atoms#: 0 (residues/phosphate atoms#: 0)
```

```
Non-protein/nucleic Atoms#: 244 (residues: 244)
```

```
Non-protein/nucleic resid values: [ CL (3), HOH (238), MG (2), NA (1) ]
```

```
Protein sequence:
```

```
MRIILLGAPGAGKGTQAQFIMEKYGIPQISTGDMRLRAAVKSGSELGKQAKDIMDAGKLV  
DELVIALVKERIAQEDCRNGFLLDGFPRTPQADAMKEAGINVDYVLEFDVPDELIVDKI  
VGRRVHAPSGRVYHVKFNPPKVEGKDDVTGEELTTRKDDQEETVRKRLVEYHQM  
TAPLIG  
YYSKEAEAGNTKYAKVDGTPVAEVRADLEKILG
```

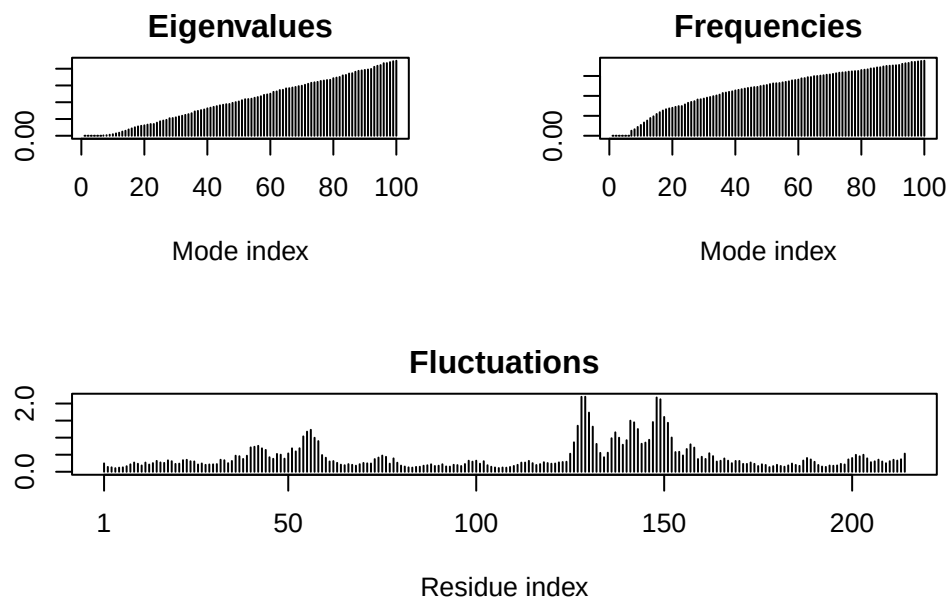
```
+ attr: atom, xyz, seqres, helix, sheet,  
      calpha, remark, call
```

```
# Perform flexibility prediction  
m <- nma(adk)
```

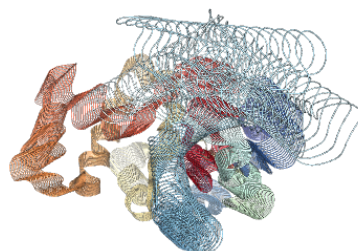
```
Building Hessian... Done in 0.03 seconds.
```

```
Diagonalizing Hessian... Done in 0.19 seconds.
```

```
plot(m)
```



```
mktrj(m, file = "adk_m7.pdb")  
  
view.nma(m, pdb=adk)
```



COMPARATIVE STRUCTURE ANALYSIS OF ADENYLATE KINASE

Q10. Which of the packages above is found only on BioConductor and not CRAN?

A10. msa

Q11. Which of the above packages is not found on BioConductor or CRAN?

A11. bio3dview

Q12. True or False? Functions from the pak package can be used to install packages from GitHub and BitBucket.

A12. TRUE

SEARCH AND RETRIEVE ADK STRUCTURES

```
library(bio3d)
aa <- get.seq("1ake_A")
```

Warning in get.seq("1ake_A"): Removing existing file: seqs.fasta

Fetching... Please wait. Done.

aa

```

      1      .      .      .      .      .      .      60
pdb|1AKE|A  MRIILLGAPGAGKGTQAQFIMEKYGIPQISTGDMRLRAAVKSGSELGKQAKDIMDAGKLV
      1      .      .      .      .      .      .      60

      61      .      .      .      .      .      .      120
pdb|1AKE|A  DELVIALVKERIAQEDCRNGFLLDGFRTIPQADAMKEAGINVDYVLEFDVPDELIVDRI
      61      .      .      .      .      .      .      120

      121      .      .      .      .      .      .      180
pdb|1AKE|A  VGRRVHAPSGRVYHVKNPPKVEGKDDVTGEELTTRKDDQEETVRKRLVEYHQMTPALIG
      121      .      .      .      .      .      .      180

      181      .      .      .      214
pdb|1AKE|A  YYSKEAEAGNTKYAKVDGTPVAEVRADLEKILG
      181      .      .      .      214
```

Call:

```
read.fasta(file = outfile)
```

Class:

```
fasta
```

Alignment dimensions:

```
1 sequence rows; 214 position columns (214 non-gap, 0 gap)
```

```
+ attr: id, ali, call
```

Q13. How many amino acids are in this sequence, i.e. how long is this sequence?

A13. 214

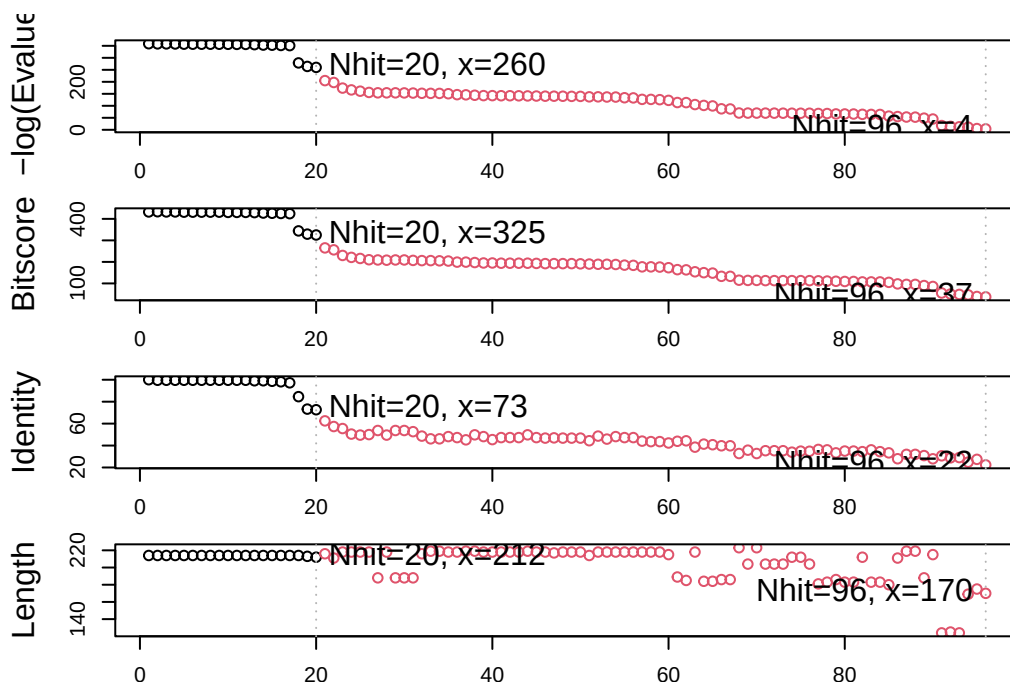
```
#BLAST search  
b <- blast.pdb(aa)
```

```
Searching ... please wait (updates every 5 seconds) RID = ZDX2BETU014  
.....  
Reporting 96 hits
```

```
hits <- plot(b)
```

```
* Possible cutoff values: 260 3  
    Yielding Nhits:      20 96
```

```
* Chosen cutoff value of: 260  
    Yielding Nhits:      20
```



```
#Top hits
head(hits$ pdb.id)
```

```
[1] "1AKE_A" "8BQF_A" "4X8M_A" "6S36_A" "9R6U_A" "9R71_A"
```

```
#"Required"
hits <- NULL
hits$ pdb.id <- c('1AKE_A','6S36_A','6RZE_A','3HPR_A','1E4V_A','5EJE_A','1E4Y_A','3X2S_A','6H
```

ANNOTATE COLLECTED PDB STRUCTURES

```
#Download related PDB files
files <- get.pdb(hits$ pdb.id, path="pdb", split = TRUE, gzip = TRUE)
```

```
Warning in get.pdb(hits$ pdb.id, path = "pdb", split = TRUE, gzip = TRUE):
pdb/1AKE.pdb exists. Skipping download
```

```
Warning in get.pdb(hits$ pdb.id, path = "pdb", split = TRUE, gzip = TRUE):
pdb/6S36.pdb exists. Skipping download
```

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/6RZE.pdb exists. Skipping download

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/3HPR.pdb exists. Skipping download

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/1E4V.pdb exists. Skipping download

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/5EJE.pdb exists. Skipping download

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/1E4Y.pdb exists. Skipping download

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/3X2S.pdb exists. Skipping download

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/6HAP.pdb exists. Skipping download

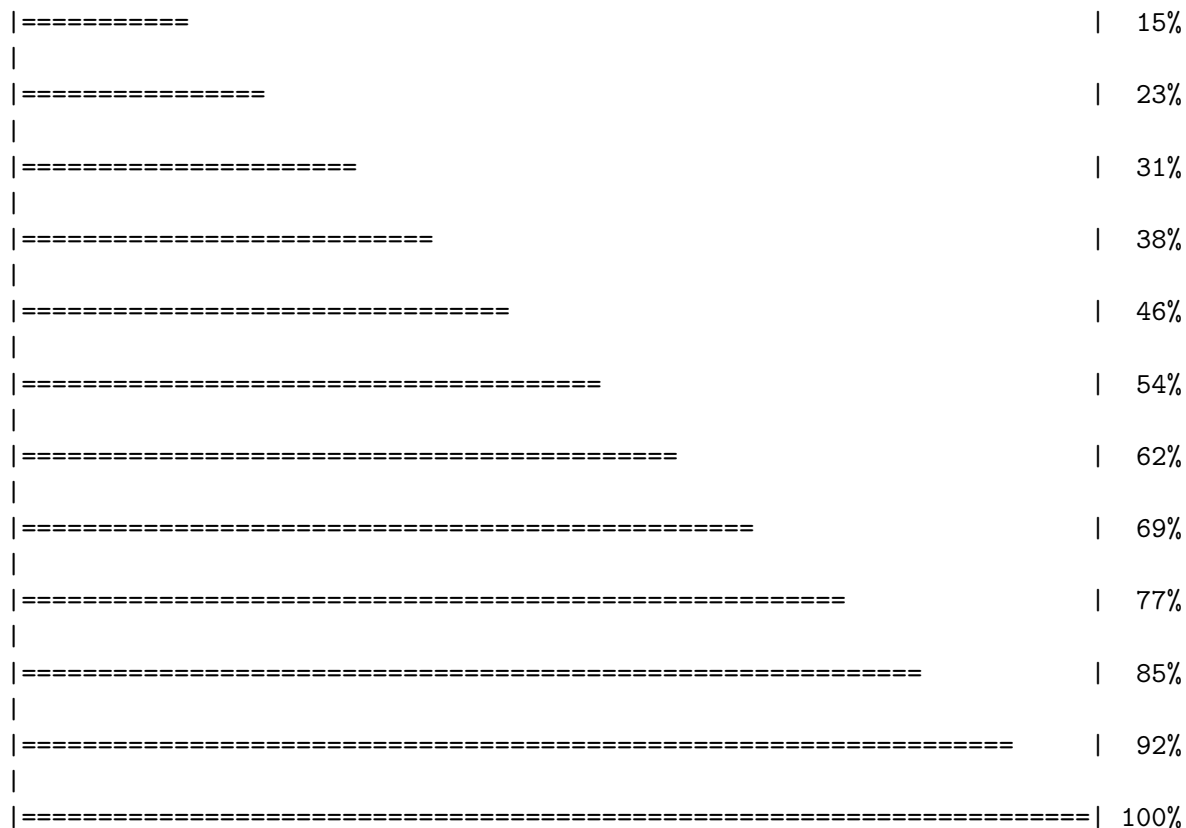
Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/6HAM.pdb exists. Skipping download

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/4K46.pdb exists. Skipping download

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/3GMT.pdb exists. Skipping download

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/4PZL.pdb exists. Skipping download

		0%
=====		8%



```
#Align related PDBs
pdbs <- pdbaln(files, fit = TRUE, exefile = "msa")
```

Reading PDB files:

```
pdbs/split_chain/1AKE_A.pdb
pdbs/split_chain/6S36_A.pdb
pdbs/split_chain/6RZE_A.pdb
pdbs/split_chain/3HPR_A.pdb
pdbs/split_chain/1E4V_A.pdb
pdbs/split_chain/5EJE_A.pdb
pdbs/split_chain/1E4Y_A.pdb
pdbs/split_chain/3X2S_A.pdb
pdbs/split_chain/6HAP_A.pdb
pdbs/split_chain/6HAM_A.pdb
pdbs/split_chain/4K46_A.pdb
pdbs/split_chain/3GMT_A.pdb
pdbs/split_chain/4PZL_A.pdb
```

PDB has ALT records, taking A only, rm.alt=TRUE

```
.   PDB has ALT records, taking A only, rm.alt=TRUE
.   PDB has ALT records, taking A only, rm.alt=TRUE
.   PDB has ALT records, taking A only, rm.alt=TRUE
..  PDB has ALT records, taking A only, rm.alt=TRUE
.... PDB has ALT records, taking A only, rm.alt=TRUE
.   PDB has ALT records, taking A only, rm.alt=TRUE
...
```

Extracting sequences

```
pdb/seq: 1   name: pdbs/split_chain/1AKE_A.pdb
           PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 2   name: pdbs/split_chain/6S36_A.pdb
           PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 3   name: pdbs/split_chain/6RZE_A.pdb
           PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 4   name: pdbs/split_chain/3HPR_A.pdb
           PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 5   name: pdbs/split_chain/1E4V_A.pdb
pdb/seq: 6   name: pdbs/split_chain/5EJE_A.pdb
           PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 7   name: pdbs/split_chain/1E4Y_A.pdb
pdb/seq: 8   name: pdbs/split_chain/3X2S_A.pdb
pdb/seq: 9   name: pdbs/split_chain/6HAP_A.pdb
pdb/seq: 10  name: pdbs/split_chain/6HAM_A.pdb
           PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 11  name: pdbs/split_chain/4K46_A.pdb
           PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 12  name: pdbs/split_chain/3GMT_A.pdb
pdb/seq: 13  name: pdbs/split_chain/4PZL_A.pdb
```

```
#Vector containing PDB database codes
ids <- basename.pdb(pdb$id)

anno <- pdb.annotate(ids)
unique(anno$source)
```

```
[1] "Escherichia coli"
[2] "Escherichia coli K-12"
[3] "Escherichia coli 0139:H28 str. E24377A"
[4] "Escherichia coli str. K-12 substr. MDS42"
[5] "Photobacterium profundum"
```

- [6] "Burkholderia pseudomallei 1710b"
 [7] "Francisella tularensis subsp. tularensis SCHU S4"

anno

	structureId	chainId	macromoleculeType	chainLength	experimentalTechnique
1AKE_A	1AKE	A	Protein	214	X-
ray					
6S36_A	6S36	A	Protein	214	X-
ray					
6RZE_A	6RZE	A	Protein	214	X-
ray					
3HPR_A	3HPR	A	Protein	214	X-
ray					
1E4V_A	1E4V	A	Protein	214	X-
ray					
5EJE_A	5EJE	A	Protein	214	X-
ray					
1E4Y_A	1E4Y	A	Protein	214	X-
ray					
3X2S_A	3X2S	A	Protein	214	X-
ray					
6HAP_A	6HAP	A	Protein	214	X-
ray					
6HAM_A	6HAM	A	Protein	214	X-
ray					
4K46_A	4K46	A	Protein	214	X-
ray					
3GMT_A	3GMT	A	Protein	230	X-
ray					
4PZL_A	4PZL	A	Protein	242	X-
ray					
	resolution	scopDomain	pfam		
1AKE_A	2.00	Adenylate kinase	Adenylate kinase, active site lid	(ADK_lid)	
6S36_A	1.60	<NA>	Adenylate kinase, active site lid	(ADK_lid)	
6RZE_A	1.69	<NA>	Adenylate kinase, active site lid	(ADK_lid)	
3HPR_A	2.00	<NA>	Adenylate kinase, active site lid	(ADK_lid)	
1E4V_A	1.85	Adenylate kinase	Adenylate kinase, active site lid	(ADK_lid)	
5EJE_A	1.90	<NA>	Adenylate kinase, active site lid	(ADK_lid)	
1E4Y_A	1.85	Adenylate kinase	Adenylate kinase, active site lid	(ADK_lid)	
3X2S_A	2.80	<NA>	<NA>	<NA>	
6HAP_A	2.70	<NA>	Adenylate kinase, active site lid	(ADK_lid)	

6HAM_A	2.55	<NA> Adenylate kinase, active site lid (ADK_lid)
4K46_A	2.01	<NA> Adenylate kinase, active site lid (ADK_lid)
3GMT_A	2.10	<NA> <NA>
4PZL_A	2.10	<NA> Adenylate kinase (ADK)
	ligandId	
1AKE_A	AP5	
6S36_A	CL (3),NA,MG (2)	
6RZE_A	NA (3),CL (2)	
3HPR_A	AP5	
1E4V_A	AP5	
5EJE_A	AP5,CO	
1E4Y_A	AP5	
3X2S_A	JPY (2),AP5,MG	
6HAP_A	AP5	
6HAM_A	AP5	
4K46_A	ADP,AMP,PO4	
3GMT_A	SO4 (2)	
4PZL_A	CA,FMT,GOL	
		ligandName
1AKE_A		BIS(ADENOSINE)-5'-
PENTAPHOSPHATE		
6S36_A		CHLORIDE ION (3),SODIUM ION,MAGNESIUM ION (2)
6RZE_A		SODIUM ION (3),CHLORIDE ION (2)
3HPR_A		BIS(ADENOSINE)-5'-
PENTAPHOSPHATE		
1E4V_A		BIS(ADENOSINE)-5'-
PENTAPHOSPHATE		
5EJE_A		BIS(ADENOSINE)-5'-PENTAPHOSPHATE,COBALT (II) ION
1E4Y_A		BIS(ADENOSINE)-5'-
PENTAPHOSPHATE		
3X2S_A		N-(pyren-1-ylmethyl)acetamide (2),BIS(ADENOSINE)-5'-PENTAPHOSPHATE,MAGNESIUM ION
6HAP_A		BIS(ADENOSINE)-5'-
PENTAPHOSPHATE		
6HAM_A		BIS(ADENOSINE)-5'-
PENTAPHOSPHATE		
4K46_A		ADENOSINE-5'-DIPHOSPHATE,ADENOSINE MONOPHOSPHATE,PHOSPHATE ION
3GMT_A		SULFATE ION (2)
4PZL_A		CALCIUM ION,FORMIC ACID,GLYCEROL
	source	
1AKE_A		Escherichia coli
6S36_A		Escherichia coli
6RZE_A		Escherichia coli
3HPR_A		Escherichia coli K-12

1E4V_A Escherichia coli
 5EJE_A Escherichia coli 0139:H28 str. E24377A
 1E4Y_A Escherichia coli
 3X2S_A Escherichia coli str. K-12 substr. MDS42
 6HAP_A Escherichia coli 0139:H28 str. E24377A
 6HAM_A Escherichia coli K-12
 4K46_A Photobacterium profundum
 3GMT_A Burkholderia pseudomallei 1710b
 4PZL_A Francisella tularensis subsp. tularensis SCHU S4

1AKE_A STRUCTURE OF THE COMPLEX BETWEEN ADENYLATE KINASE FROM ESCHERICHIA COLI AND THE INHIB
 6S36_A
 6RZE_A
 3HPR_A
 1E4V_A
 loop
 5EJE_A
 1E4Y_A
 loop
 3X2S_A
 conjugated adenylate kinase
 6HAP_A
 6HAM_A
 4K46_A
 3GMT_A
 4PZL_A

Cryst

The crys

		citation	rObserved	rFree
1AKE_A	Muller, C.W., et al.	J Mol Biology (1992)	0.19600	NA
6S36_A	Rogne, P., et al.	Biochemistry (2019)	0.16320	0.23560
6RZE_A	Rogne, P., et al.	Biochemistry (2019)	0.18650	0.23500
3HPR_A	Schrank, T.P., et al.	Proc Natl Acad Sci U S A (2009)	0.21000	0.24320
1E4V_A	Muller, C.W., et al.	Proteins (1993)	0.19600	NA
5EJE_A	Kovermann, M., et al.	Proc Natl Acad Sci U S A (2017)	0.18890	0.23580
1E4Y_A	Muller, C.W., et al.	Proteins (1993)	0.17800	NA
3X2S_A	Fujii, A., et al.	Bioconjug Chem (2015)	0.20700	0.25600
6HAP_A	Kantaev, R., et al.	J Phys Chem B (2018)	0.22630	0.27760
6HAM_A	Kantaev, R., et al.	J Phys Chem B (2018)	0.20511	0.24325
4K46_A	Cho, Y.-J., et al.	To be published	0.17000	0.22290
3GMT_A	Buchko, G.W., et al.	Biochem Biophys Res Commun (2010)	0.23800	0.29500
4PZL_A	Tan, K., et al.	To be published	0.19360	0.23680

rWork spaceGroup

1AKE_A 0.19600 P 21 2 21

6S36_A 0.15940 C 1 2 1

```

6RZE_A 0.18190    C 1 2 1
3HPR_A 0.20620    P 21 21 2
1E4V_A 0.19600    P 21 2 21
5EJE_A 0.18630    P 21 2 21
1E4Y_A 0.17800    P 1 21 1
3X2S_A 0.20700    P 21 21 21
6HAP_A 0.22370    I 2 2 2
6HAM_A 0.20311      P 43
4K46_A 0.16730    P 21 21 21
3GMT_A 0.23500    P 1 21 1
4PZL_A 0.19130      P 32

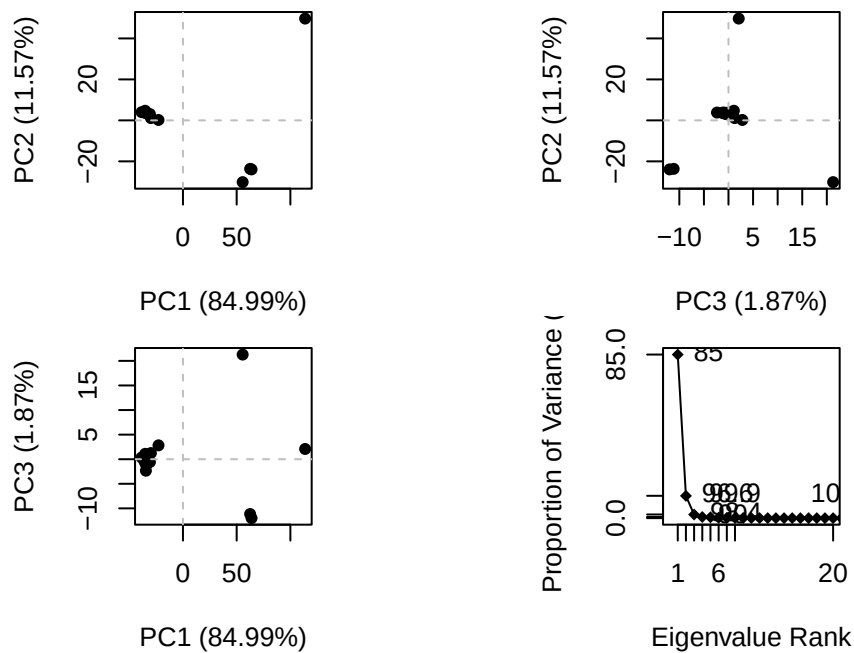
```

PRINCIPAL COMPONENT ANALYSIS

```

# Perform PCA
pc.xray <- pca(pdbx)
plot(pc.xray)

```



```

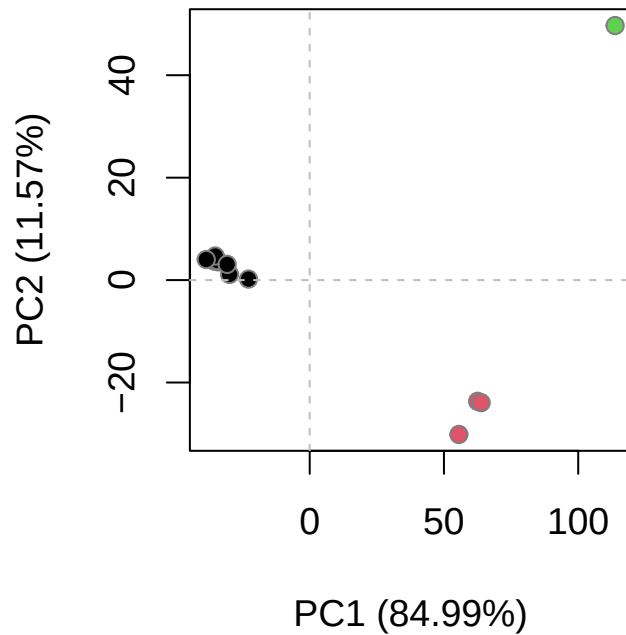
#Calculate RMSD
rd <- rmsd(pdbx)

```

Warning in rmsd(pdbx): No indices provided, using the 204 non NA positions

```
#Structure-based clustering
hc.rd <- hclust(dist(rd))
grps.rd <- cutree(hc.rd, k=3)

plot(pc.xray, 1:2, col="grey50", bg=grps.rd, pch=21, cex=1)
```



PCA VISUALIZATION

```
#Visualize first principal component
pcl <- mktrj(pc.xray, pc=1, file = "pc_1.pdb")
```

```
#PCA results with ggplot
library(ggplot2)
library(ggrepel)

df <- data.frame(PC1=pc.xray$z[,1],
                 PC2=pc.xray$z[,2],
                 col=as.factor(grps.rd),
                 ids=ids)

p <- ggplot(df) +
  aes(PC1, PC2, col=col, label=ids) +
```

```
geom_point(size=2) +
geom_text_repel(max.overlaps = 20) +
theme(legend.position = "none")
```

p

